

$$g = \frac{1}{|B|} \sum_{i \in B} c_i g_i + \sigma \mathcal{N}(0, I_d) \in \mathbb{R}^d$$

$$\approx \frac{c \sum g_i + \sigma \mathcal{N}(0, I_d)}{B}$$

param update: $w_{t+1} = w_t - \eta g_t$

$|B|$ = Batch size
 σ = scale of DP noise.
 $c_i = \min(1, \frac{c}{\|g_i\|})$
 d = # of param.

$$f(x + \Delta x) \approx f(x) + f'(x)(\Delta x) + \frac{1}{2} f''(x)(\Delta x)^2$$

$$L(w - \eta g) \approx L(w) - \eta G^T g + \frac{\eta^2}{2} g^T H g$$

$$G = \nabla L(w) \in \mathbb{R}^d \Rightarrow \frac{L(w) - L(w - \eta g)}{\Delta L} = -\frac{\eta}{2} g^T H g + \eta G^T g$$

$$H = \nabla^2 L(w) \in \mathbb{R}^{d \times d}$$

Curvature penalty

① individual gradients follow iid

$$E[g_i] = G \quad \text{Cov}(g_i) = \left(\sum \right)$$

$$E[g] = \frac{c}{B} \sum E[g_i] + \frac{\sigma}{B} E[\tilde{z}]$$

$$= \frac{c}{B} B \cdot G + 0 = cG$$

$$E[g] = cG$$

$$\text{Cov}(g_i) = \frac{c^2}{B} \Sigma + \frac{\sigma^2}{B^2} I$$

$$\text{tr}(A+B) = \text{tr}(A) + \text{tr}(B)$$

$$\text{tr}(a) = a$$

$$\text{tr}(AB) = \text{tr}(BA)$$

$$\Delta \mathcal{L} = \eta G^T E[g] - \frac{\eta^2}{2} E[g^T H g]$$

$$= \eta C G^T G - \frac{\eta^2}{2} (\text{tr}(H \text{Cov}(g)) + E[g]^T H E[g])$$

$$\Delta \mathcal{L} = \eta C G^T G - \frac{\eta^2}{2} \left(C^T G^T H G + \frac{C^T \text{tr}(H \Sigma)}{B^T} + \frac{\eta^2 \text{tr}(H)}{B^T} \right)$$

$C =$

$$\text{tr}(H) = \lambda_1 + \lambda_2 + \dots \rightarrow$$